MPG No. 12.3.42 4′22′ (4’22’ setting) [Type III, tetragonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & -M_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{xx} = \mathbb{M}_{v}^{(1)}$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$M_{yzx} = \mathbb{M}_{xyz}^{(1)}$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & -2M_{yzx} \end{bmatrix}$$

$M_{yzx} = \mathbb{T}_{v}^{(1)}$
$2M_{yzx} = 2\mathbb{T}_{v}^{(1)}$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{yzx} = \mathbb{T}_{v}^{(2)}$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxzz} & 0 & 0 & 0 \\ 0 & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 \\ M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$M_{xxxx} = -\mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_{v}^{(1)} + 4\mathbb{M}_{v}^{(2)}$
$M_{xxzz} = \mathbb{M}_{4v}^{(1)} + \mathbb{M}_{v}^{(1)}$
$M_{yzyz} = -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_{v}^{(2)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 \\ -M_{xxyy} & 0 & -M_{xxzz} & 0 & 0 & 0 \\ -M_{xxzz} & M_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$
$$M_{xxzz} = -\mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & 0 & 0 \\ 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = 2\mathbb{M}_v^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzyz} & 0 & 0 \\ 0 & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$
$$M_{xyxy} = 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$
$$M_{xyxy} = 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyy} & -M_{xxxx} & -M_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{zzxx} & -M_{zzxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & -M_{yzzy} & 0 & 0 & -M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = -\mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$
$$M_{xxyy} = -2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$
$$M_{xxzz} = \mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$
$$M_{zzxx} = \mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$
$$M_{yzyz} = -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$
$$M_{yzzy} = -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$
$$M_{xyxy} = 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & 0 & M_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzy} & 0 & 0 & M_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$
\begin{aligned}
M_{yzyz} &= 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzzy} &= -2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{xyxy} &= 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
M_{xxxx} & 0 & 0 \\
0 & -M_{xxxx} & 0 \\
0 & 0 & 0 \\
0 & 0 & M_{yyzz} \\
M_{zzxx} & 0 & 0 \\
0 & -M_{yyzz} - M_{zzxx} & 0 \\
0 & M_{yzzy} & 0 \\
0 & 0 & M_{zxxz} \\
-M_{yzzy} - M_{zxxz} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxxx} &= -\mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)} \\
M_{yyzz} &= -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{zzxx} &= \mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{yzzy} &= -\mathbb{M}_{4v}^{(1)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \\
M_{zxxz} &= \mathbb{M}_{4v}^{(1)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)}
\end{aligned}
$$